

COOPERATIVE NODE-SPECIFIC SPEECH ENHANCEMENT IN RESOURCE-CONSTRAINED ACOUSTIC SENSOR NETWORKS

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ABSTRACT

We study node-specific speech enhancement in a distributed acoustic sensor network with one microphone per node, where each node retains its local talkers while suppressing competing speech, music, and background noise. We compare local-only and graph-conditioned models using matched backbone architectures under different communication regimes. Local processing already provides strong gains under favorable conditions, but degrades in low-SNR, high-interference scenarios. In these regimes, cross-node cooperation improves SI-SDR by up to 4.8 dB over the matched local model. We further characterize the performance-bandwidth tradeoff: low-rate bottleneck sharing provides moderate gains, whereas full feature sharing achieves the best performance at substantially higher communication cost.

Index Terms— Distributed speech enhancement, acoustic sensor networks, graph-conditioned learning

1 Introduction

Distributed acoustic sensor networks are increasingly considered for conversational environments in which several groups of nearby talkers occupy the same space, such as adjacent tables in restaurants, meeting areas, or social venues. In these settings, each node captures a mixture dominated by its local talkers together with interfering speech from other groups, background music, and noise. The objective is node-specific speech enhancement: each node must retain only its local talkers while suppressing all other sources, so that different nodes pursue different enhancement targets while exchanging information. Such systems are relevant to assistive listening, teleconferencing, and smart audio capture in shared spaces [1–3].

This problem is particularly challenging in the single-microphone-per-node regime. Unlike centralized or local multi-microphone systems, spatial filtering methods such as MVDR, LCMV, and multichannel Wiener filtering cannot be applied independently at each node [4–6]. Distributed

formulations, including DANSE and related approaches, extend linear spatial filtering through inter-node communication [1–3], but remain based primarily on linear estimation and second-order signal statistics.

In contrast, supervised single-channel speech enhancement based on time–frequency masking and encoder–decoder architectures has shown strong performance under challenging conditions [7–11]. These models can capture nonstationary and non-Gaussian speech structure, but typically operate independently at each node and do not exploit observations from the rest of the network. Moreover, they do not directly address the key difficulty in this setting: each node has a different target signal, and the target of one node acts as interference at the others.

Recent graph-based and topology-aware approaches enable cross-node information exchange through feature aggregation and attention mechanisms [12–15]. Although these methods demonstrate the value of graph structure for multi-channel and ad hoc array processing, it remains unclear when communication is necessary in the more constrained single-microphone-per-node setting and whether the resulting gains justify the added cost in battery-powered and bandwidth-limited systems. Practical networks may also exhibit timing and sampling-rate offsets across nodes [16, 17], which motivates communication schemes that do not depend on cross-node phase alignment.

In this work, we compare local and cooperative models using matched backbone architectures under three communication regimes. The *Local* model uses only node-wise observations, the *Neighborhood* model exchanges information with adjacent nodes, and the *Full network* model uses information from all nodes. We evaluate these models across acoustic conditions of increasing difficulty.

Our results show that local processing already performs strongly under favorable conditions. As the acoustic conditions become more adverse, however, the benefit of cross-node cooperation increases substantially. We also observe a clear performance–bandwidth tradeoff: low-rate bottleneck exchange provides moderate gains, whereas full feature sharing achieves the best performance at higher communication cost.

Contributions: (i) We formulate node-specific speech enhancement in a distributed network with one microphone

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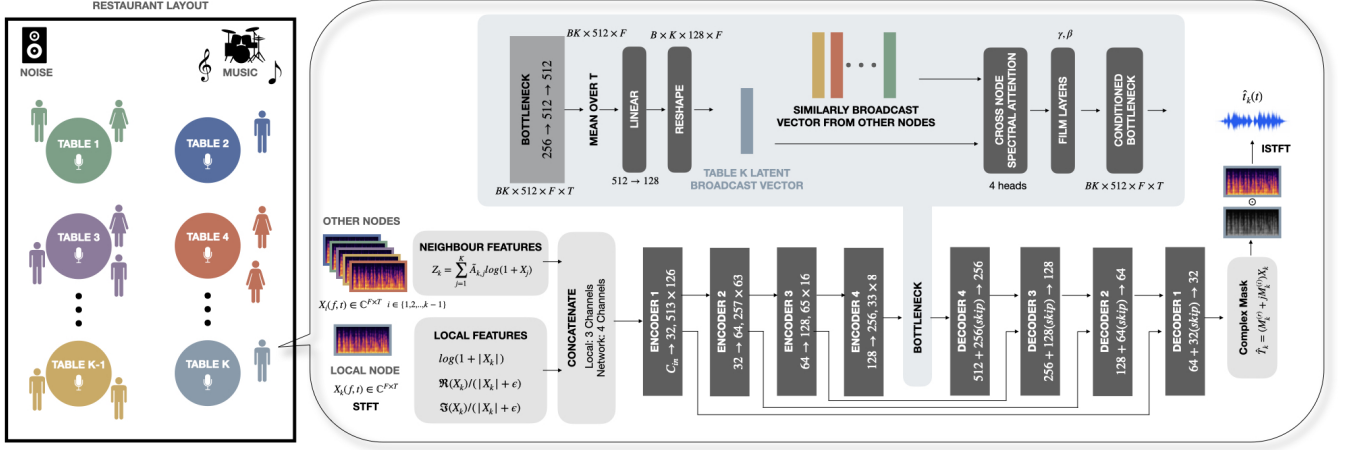


Fig. 1: Left: Distributed conversational scenario with multiple groups of talkers. Each node observes local and interfering speech and has a node-specific target. Right: Proposed model. Neighbor log-magnitude features are shared at the input, and cross-node attention is applied at the bottleneck. The network outputs a node-specific mask.

per node. (ii) We compare local and graph-conditioned models using matched backbone architectures under different communication regimes. (iii) We show that the benefit of cross-node cooperation is condition-dependent and quantify the corresponding performance–bandwidth tradeoff.

2 Method

Problem Formulation. We consider a distributed acoustic sensor network with K nodes, each equipped with a single microphone. Let $\mathcal{S}_k$ denote the set of local talkers associated with node k , and let $s_{i \rightarrow k}[n]$ denote the reverberant contribution of talker i at node k . We define

$$t_k[n] = \sum_{i \in \mathcal{S}_k} s_{i \rightarrow k}[n] \quad (1)$$

as the local target speech at node k , with STFT $T_k(f, t)$. Each node observes local talkers together with interfering talkers from other nodes, background music, and noise. The observed mixture is

$$X_k(f, t) = T_k(f, t) + \sum_{j \neq k} I_{j \rightarrow k}(f, t) + N_k(f, t), \quad (2)$$

where $I_{j \rightarrow k}$ denotes the summed reverberant contribution from talkers near node j as observed at node k , and N_k includes background noise and music. The goal is to estimate $\hat{T}_k(f, t)$, where the target at one node acts as interference at others. We assume frame-synchronous nodes, but avoid cross-node phase because sampling-rate offsets or motion may make it unreliable [16, 17].

Communication Regimes. We compare three communication regimes built on the same encoder–decoder backbone: *Local*, *Neighborhood*, and *Full network*. The *Local*

model uses only the observation at node k . The *Neighborhood* model exchanges information with adjacent nodes defined by a fixed graph, while the *Full network* model uses information from all nodes.

Each model estimates a T-F mask from the mixture representation. Unless otherwise noted, we use a complex ratio mask (CRM); other mask formulations showed similar trends and are omitted for brevity.

Local and Shared Features. Given the complex short-time Fourier transform X_k , local features are

$$[\log(1 + |X_k|), \Re(X_k)/(|X_k| + \epsilon), \Im(X_k)/(|X_k| + \epsilon)], \quad (3)$$

which provide magnitude and normalized phase information.

To incorporate cross-node information, we aggregate neighbor log-magnitude spectra using an adjacency matrix $A \in \mathbb{R}^{K \times K}$. Self-connections are removed and each row is normalized to sum to one, yielding $\tilde{A}$. We then compute

$$Z_k(f, t) = \sum_{j=1}^K \tilde{A}_{k,j} \log(1 + |X_j(f, t)|). \quad (4)$$

Only log-magnitude is shared because phase is not aligned across nodes. Cooperative models concatenate Z_k with the local features, while the *Local* model uses only local features.

Network Architecture. The proposed model is illustrated in Fig. 1. The backbone is a U-Net with convolutional encoder–decoder blocks and skip connections, shared across all regimes. Cross-node information is integrated at two stages: neighbor log-magnitude features at the input and cross-node attention at the bottleneck. The resulting graph context conditions local bottleneck and decoder features using FiLM,

$$\text{FiLM}(H_k, c_k) = (1 + \gamma_k) \odot H_k + \beta_k, \quad (5)$$

where (γ_k, β_k) are predicted from conditioning vector c_k . The *Local* model has 8.55M parameters, while the two cooperative models have 8.83M parameters.

Mask Estimation. The network estimates real and imaginary components $M_k^{(r)}(f, t)$ and $M_k^{(i)}(f, t)$, and reconstructs

$$\hat{T}_k(f, t) = (M_k^{(r)}(f, t) + j M_k^{(i)}(f, t)) X_k(f, t). \quad (6)$$

Training Objective. All models are trained using the waveform-domain SI-SDR loss [18]:

$$\mathcal{L} = -\text{SI-SDR}(\hat{t}_k, t_k). \quad (7)$$

3 Experiments

We simulate a distributed restaurant-style acoustic sensor network with $K = 4$ nodes (tables), each equipped with a single microphone. The room measures $12 \times 10 \times 3$ m, with table centers at (3, 3), (9, 3), (3, 7), and (9, 7) m. Speakers are placed 0.4–0.6 m from their associated table center. Each node has a node-specific target consisting of its local talkers, while speech from other tables, background music, and ambient noise act as interference. Clean speech is drawn from LibriSpeech [19] and VCTK [20], and noise and music from MUSAN [21]. Microphone signals are generated using convolutive mixing with simulated room impulse responses, consistent with (2).

All scenes are 5 s long at 16 kHz. We use 5000 training and 500 validation scenes from the *Low-noise* regime, and three disjoint 500-scene test sets: *Low-noise* (SNR 5–20 dB, RT60 0.3–0.6 s, 1–3 talkers/table), *Medium-noise* (SNR 0–8 dB, RT60 0.5–0.8 s, 2–3 talkers/table), and *High-noise* (SNR –8–0 dB, RT60 0.6–0.9 s, 3 talkers/table). The *Neighborhood* model connects each node to its nearest table, while the *Full network* model uses an all-to-all topology. We additionally report a $K = 6$ ablation.

All models use the same U-Net backbone and are trained under identical conditions. Performance is evaluated using SI-SDR, PESQ, and STOI, averaged over nodes and scenes. We report results across acoustic regimes and define *cooperation gain* as the SI-SDR improvement of a cooperative model over its matched local baseline.

4 Results

Table 1 reports results for the selected CRM model family. The local model improves SI-SDR from 5.50 to 11.43 dB under low-noise conditions, with further improvement to 13.54 dB using the full network.

As acoustic conditions become more adverse, the benefit of cooperation increases. On the high-noise split, SI-SDR improves from 2.45 dB (local) to 7.28 dB (full network), corresponding to a 4.83 dB cooperation gain. PESQ and STOI follow the same ordering.

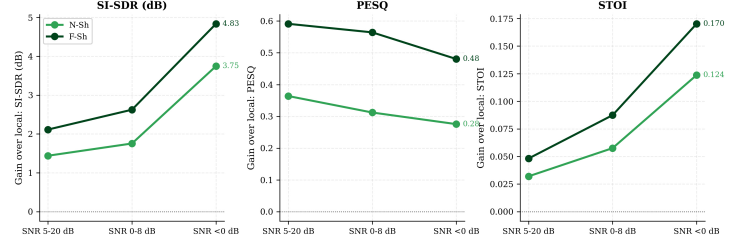


Fig. 2: Cooperation gain over the Local model across acoustic regimes. Gains increase from low- to medium- to high-noise conditions.

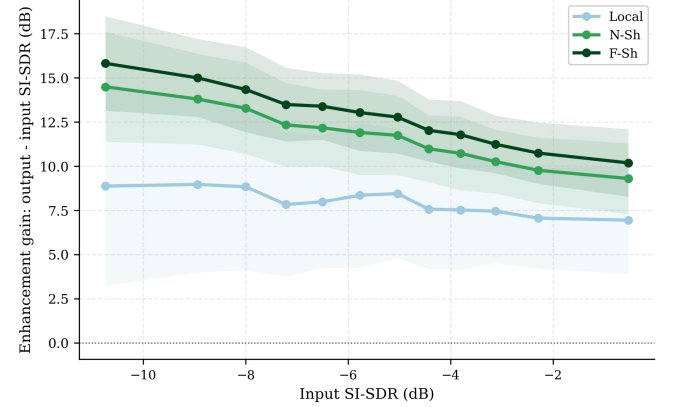


Fig. 3: Enhancement gain versus input SI-SDR on the high-noise split. Cooperative-model gains increase as input quality decreases.

Figure 2 shows cooperation gain across regimes. The gain increases from 2.11 dB (low-noise) to 2.62 dB (medium-noise) and 4.83 dB (high-noise), indicating that cross-node information becomes more useful as interference increases.

Figure 3 shows per-scene results on the high-noise split. All models provide positive enhancement gains, while cooperative models maintain a larger margin over the local model across input conditions, with the largest differences at low input SI-SDR.

The full-network gain increases from 2.08 dB for one talker to 3.76 dB for three talkers. Since speaker count, SNR, and reverberation are coupled, this reflects increasing interference rather than speaker count alone. The SNR and RT60 breakdowns follow the same pattern, with larger gains in low-SNR and more reverberant conditions.

4.1 Performance–Bandwidth Tradeoff

Figure 4 shows the communication tradeoff. The local model requires no communication, while bottleneck sharing provides a low-rate operating point. Neighborhood and full-network sharing require higher bandwidth and yield larger gains, especially in the high-noise regime.

Table 1: Mean SI-SDR, PESQ, and STOI across acoustic conditions for the selected CRM model family. “In” denotes the input mixture. “N-Sh” and “F-Sh” denote the Neighborhood-shared and Full-shared models, respectively. “Gain” is F-Sh – Local in SI-SDR. Difficulty levels correspond to increasing acoustic challenge (lower SNR, more interference, and higher reverberation).

Setting	Condition	SI-SDR (dB)					PESQ				STOI			
		In	Local	N-Sh	F-Sh	Gain	In	Local	N-Sh	F-Sh	In	Local	N-Sh	F-Sh
Overall	Low difficulty	5.50	11.43	12.87	13.54	+2.11	1.29	2.01	2.37	2.60	0.75	0.84	0.87	0.89
	Medium difficulty	0.80	7.35	9.11	9.98	+2.62	1.16	1.59	1.91	2.16	0.66	0.79	0.84	0.87
	High difficulty	-5.54	2.45	6.19	7.28	+4.83	1.12	1.29	1.57	1.77	0.54	0.65	0.78	0.82
Speakers	1 talker	3.10	11.40	12.78	13.48	+2.08	1.19	1.84	2.07	2.24	0.65	0.78	0.81	0.83
	2 talkers	2.28	8.79	10.41	11.22	+2.43	1.18	1.71	2.02	2.25	0.68	0.80	0.85	0.88
	3 talkers	-1.23	5.46	8.28	9.22	+3.76	1.19	1.56	1.90	2.13	0.64	0.74	0.83	0.86
SNR	High SNR	6.42	11.99	13.33	13.96	+1.97	1.33	2.09	2.44	2.67	0.76	0.85	0.88	0.90
	Medium SNR	1.52	8.09	9.82	10.66	+2.57	1.17	1.66	1.99	2.24	0.68	0.79	0.85	0.88
	Low SNR	-5.54	2.45	6.19	7.28	+4.83	1.12	1.29	1.57	1.77	0.54	0.65	0.78	0.82
RT60	Low RT60	6.02	11.89	13.38	14.04	+2.15	1.31	2.05	2.44	2.66	0.76	0.85	0.88	0.90
	Medium RT60	0.07	6.95	9.21	10.09	+3.14	1.17	1.60	1.92	2.16	0.65	0.77	0.84	0.87
	High RT60	-3.42	3.97	6.92	7.93	+3.97	1.13	1.38	1.65	1.87	0.58	0.69	0.79	0.83

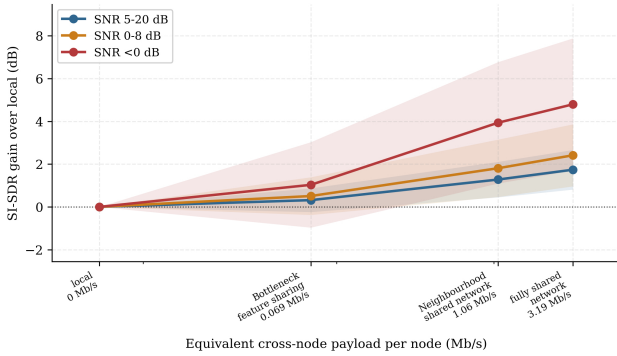


Fig. 4: SI-SDR gain versus communication bandwidth. Higher communication enables larger gains, especially in adverse conditions.

Table 2: Comparison with classical Wiener filtering methods using oracle spatial covariance matrices (SCMs).

Method	SI-SDR (dB)	Δ SI-SDR
Input mixture	5.50	–
Local MWF	7.34	+1.84
Centralized MWF (oracle)	8.70	+3.20
DANSE (oracle SCMs)	8.70	+3.20
dMWF (oracle SCMs)	8.70	+3.20
Local model (CRM)	11.43	+5.93
Full-shared model (CRM)	13.54	+8.04

At run time, input-level sharing transmits $257 \times 251 = 64,507$ float32 values per 2 s crop (258 kB, 1.03 Mb/s per stream). Bottleneck sharing transmits $128 \times 17 = 2,176$ values (8.7 kB, 34.8 kb/s per stream). For $K = 4$, the neighborhood model has degree 1 and the full-network model has degree 3, resulting in the per-node bandwidths shown in Fig. 4. All experiments assume frame-synchronous nodes.

4.2 Additional Analysis

Table 2 compares the learned models with classical Wiener filtering using oracle spatial covariance matrices. Centralized MWF, DANSE, and dMWF achieve the same SI-SDR (8.70 dB), confirming that the distributed implementations re-

cover the centralized LMMSE solution under oracle conditions. The local model exceeds this baseline under low-noise conditions (11.43 dB vs. 8.70 dB), and the full-network model improves further (13.54 dB). Since these baselines use oracle SCMs, they represent optimistic references for linear methods.

For $K = 6$ nodes, the same ordering is observed: 4.13, 5.42, and 6.27 dB for Local, Neighborhood, and Full network, corresponding to gains of 1.30 and 2.15 dB over local processing. The trend is consistent with larger gains in lower-SNR conditions.

An ablation shows that bottleneck-only sharing yields gains of +0.78 to +1.03 dB on the high-noise split, while input-level sharing provides +3.93 dB, with the full model reaching +4.80 dB. This indicates that most of the gain arises from early cross-node spectral information, with bottleneck conditioning providing additional refinement.

5 Conclusion

This work shows that the benefit of cross-node communication in distributed speech enhancement depends strongly on acoustic conditions. When local observations are informative, single-node processing is already effective and additional communication provides limited improvement. As interference increases, however, shared information across nodes provides substantial gains and becomes increasingly important for maintaining performance.

The results further show that combining nonlinear spectral modeling with cross-node information can outperform the tested local models and oracle linear spatial filtering. In this setting, cooperation exploits complementary information across nodes that is unavailable from any single observation.

These findings motivate distributed systems that adapt the amount of shared information to acoustic conditions, increasing cooperation in challenging environments while limiting communication when local processing is sufficient.

6 References

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